

# Austro Energy Systems Int. AG

Gas reformer AES3000



## Gas reformer AES3000

- Purification of associated gas from the hydrogen sulfide and high hydrocarbons and conversion of its qualities into similar properties of natural gas.
- Use of gas reformer AES improves increases performance of AES gas engines for work on associated and other gases with low methane number.
- Volume of gas preparation is 3000 Nm<sup>3</sup> / hour (the volume of prepared gas of 5500 Nm<sup>3</sup> / hour sufficient to generate min 18 MW of electricity per hour)



## Container block of gas reformer AES3000



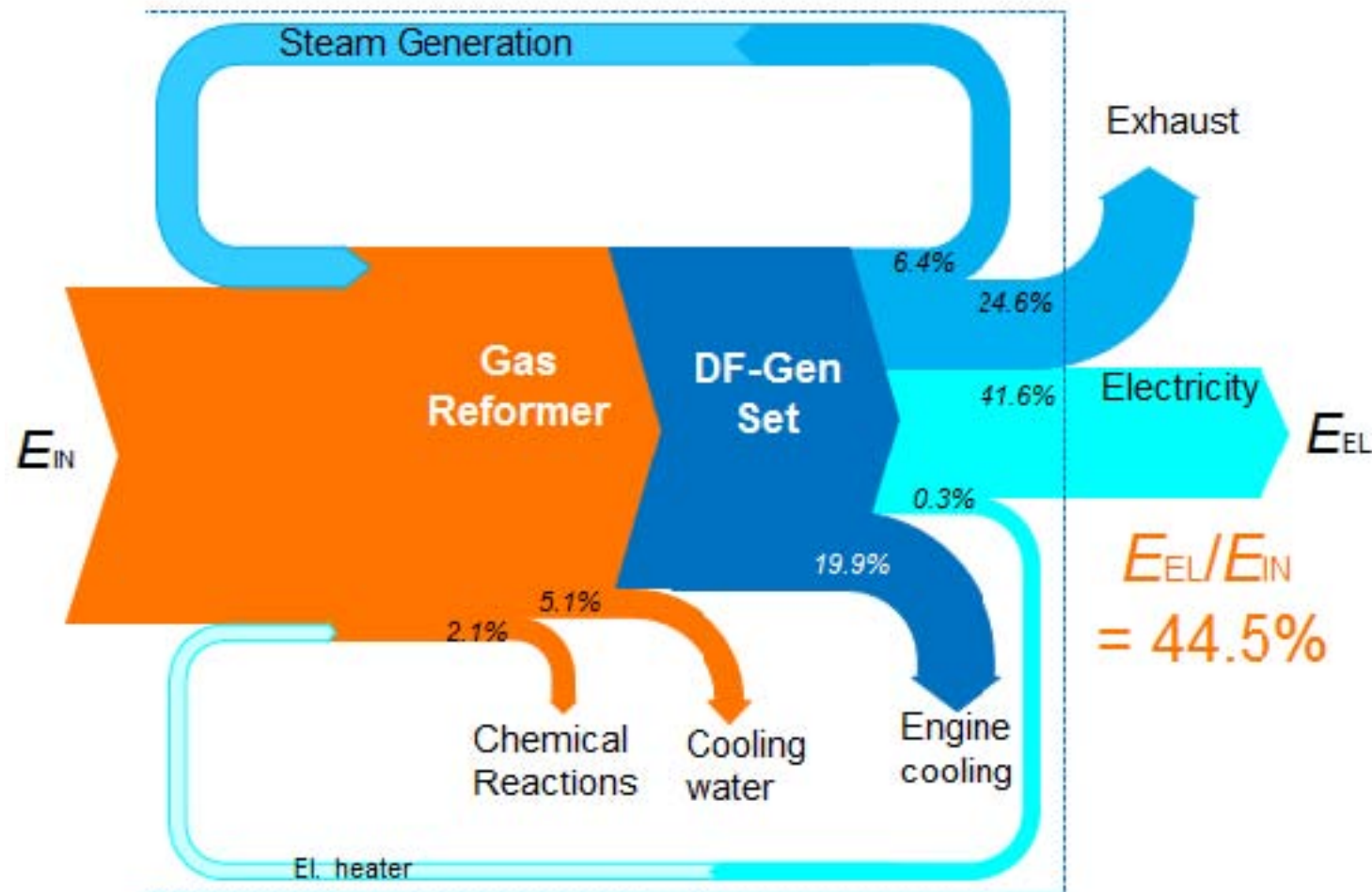
installation length— 700 mm

installation width— 4800 mm

installation height— 4200 mm

installation weight in container – 40000 kg

## Energy balance scheme gas reformer AES3000



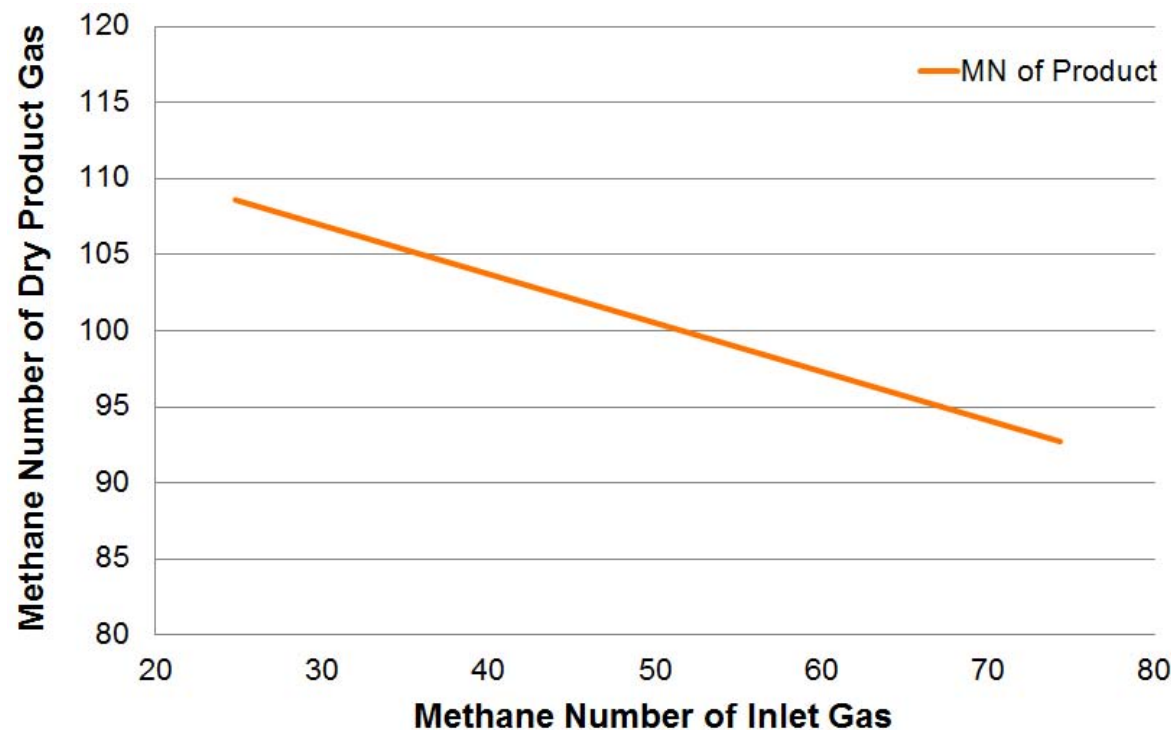
## Methane number

- Compressibility of the gas is measured by "methane number." The higher methane number, the better the fuel gas compressibility.
- Pure methane number is 100, but the presence of higher hydrocarbons in the gas significantly reduces this number.
- When the methane number of the fuel gas is 80 or lower, the engine must be used with a reduced nominal power in order to prevent knocking and engine damage.
- At high methane number, on the other hand, the engine can be operated with the highest efficiency.
- Methane number of individual components:

| Composition | Meth. number | Composition      | Meth. number | Composition      | Meth. number |
|-------------|--------------|------------------|--------------|------------------|--------------|
| Methane     | 100          | I-pentane        | 10           | H <sub>2</sub> S | 10           |
| Ethane      | 43           | N-gexane         | 10           | H <sub>2</sub>   | 0            |
| Propane     | 34           | N-septane        | 10           | N <sub>2</sub>   | 200          |
| N-butane    | 10           | H <sub>2</sub> O | 100          | O <sub>2</sub>   | 200          |
| I-butane    | 10           | CO               | 73           | Argon            | 200          |
| N-pentane   | 10           | CO <sub>2</sub>  | 200          | Helium           | 200          |

## Methane number after installation of gas reformer AES3000

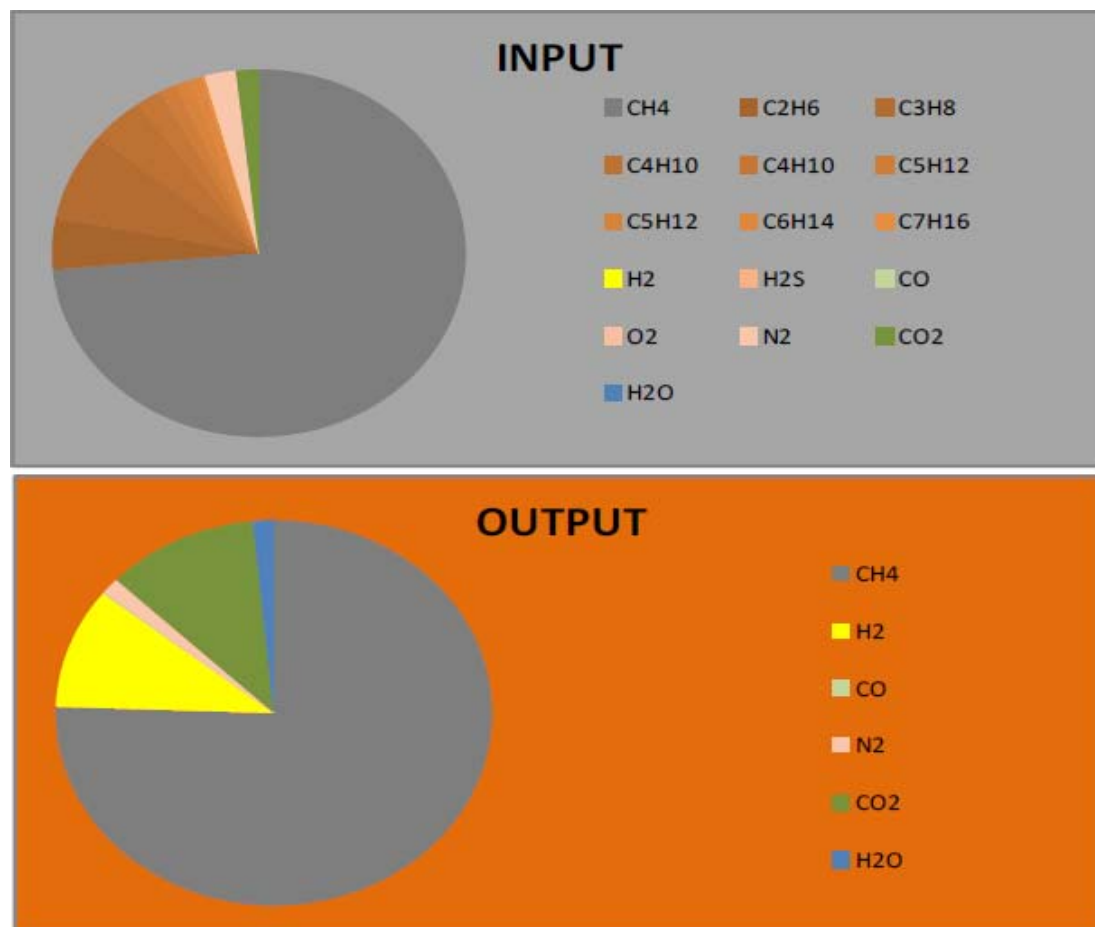
- In the GasReformer the MN of any fuel gas is improved up to  $100 \pm 5$  by converting the non-methane hydrocarbons (NMHCs) to synthesis gas ( $H_2 + CO$ ) and finally to methane ( $CH_4$ ).
- The GasReformer process is based on known steam reforming technology, where the hydrocarbons (alkanes) are reacted with steam in optimized conditions in the presence of a catalyst. This Ni-based catalyst is very sensitive to sulfur compounds, which need to be removed from the gas with a desulfurization absorbent upstream the reformer. Excess steam is separated from the product gas by condensing and recycled back to steam generation thus minimizing the need of fresh water. The dry product gas ( $50^\circ C$ ) is then fed through the Gas Valve Unit (GVU) to the Dual Fuel engine.



## Example of increasing methane number of associated gas via gas reformer AES3000

| Substance            | Formula                                          | Inlet, mol % | Outlet, mol % |
|----------------------|--------------------------------------------------|--------------|---------------|
| methane              | CH <sub>4</sub>                                  | 73,6         | 75,5          |
| ethane               | C <sub>2</sub> H <sub>6</sub>                    | 4,5          | 0,0           |
| Propane              | C <sub>3</sub> H <sub>8</sub>                    | 7,8          | 0,0           |
| n-butane             | C <sub>4</sub> H <sub>10</sub>                   | 4,2          | 0,0           |
| i-butane             | C <sub>4</sub> H <sub>10</sub>                   | 2,0          | 0,0           |
| n-pentane            | C <sub>5</sub> H <sub>12</sub>                   | 1,3          | 0,0           |
| i-pentane            | C <sub>5</sub> H <sub>12</sub>                   | 1,2          | 0,0           |
| n-hexane             | C <sub>6</sub> H <sub>14</sub>                   | 0,9          | 0,0           |
| n-heptane            | C <sub>7</sub> H <sub>16</sub>                   | 0,3          | 0,0           |
| n-oktane             | C <sub>8</sub> H <sub>18</sub>                   | 0,0          | 0,0           |
| n-nonane             | C <sub>9</sub> H <sub>20</sub>                   | 0,0          | 0,0           |
| n-dekane up to C20   | C <sub>10</sub> H <sub>22</sub> -C <sub>20</sub> | 0,0          | 0,0           |
| hydrogen             | H <sub>2</sub>                                   | 0,0          | 10,1          |
| Hydrogen sulfide     | H <sub>2</sub> S                                 | 0,0          | 0,0           |
| Carbon monoxide      | CO                                               | 0,0          | 0,1           |
| Oxygen               | O <sub>2</sub>                                   | 0,0          | 0,0           |
| nitrogen             | N <sub>2</sub>                                   | 2,4          | 1,4           |
| Carbon dioxide       | CO <sub>2</sub>                                  | 1,8          | 11,3          |
| water                | H <sub>2</sub> O                                 | 0,0          | 1,6           |
| helium               | He                                               | 0,0          | 0,0           |
| argon                | Ar                                               | 0,0          | 0,0           |
| <b>Total</b>         |                                                  | <b>100</b>   | <b>100</b>    |
| <b>Methan number</b> |                                                  | <b>42</b>    | <b>100</b>    |

## Parameters of gas reformer AES3000



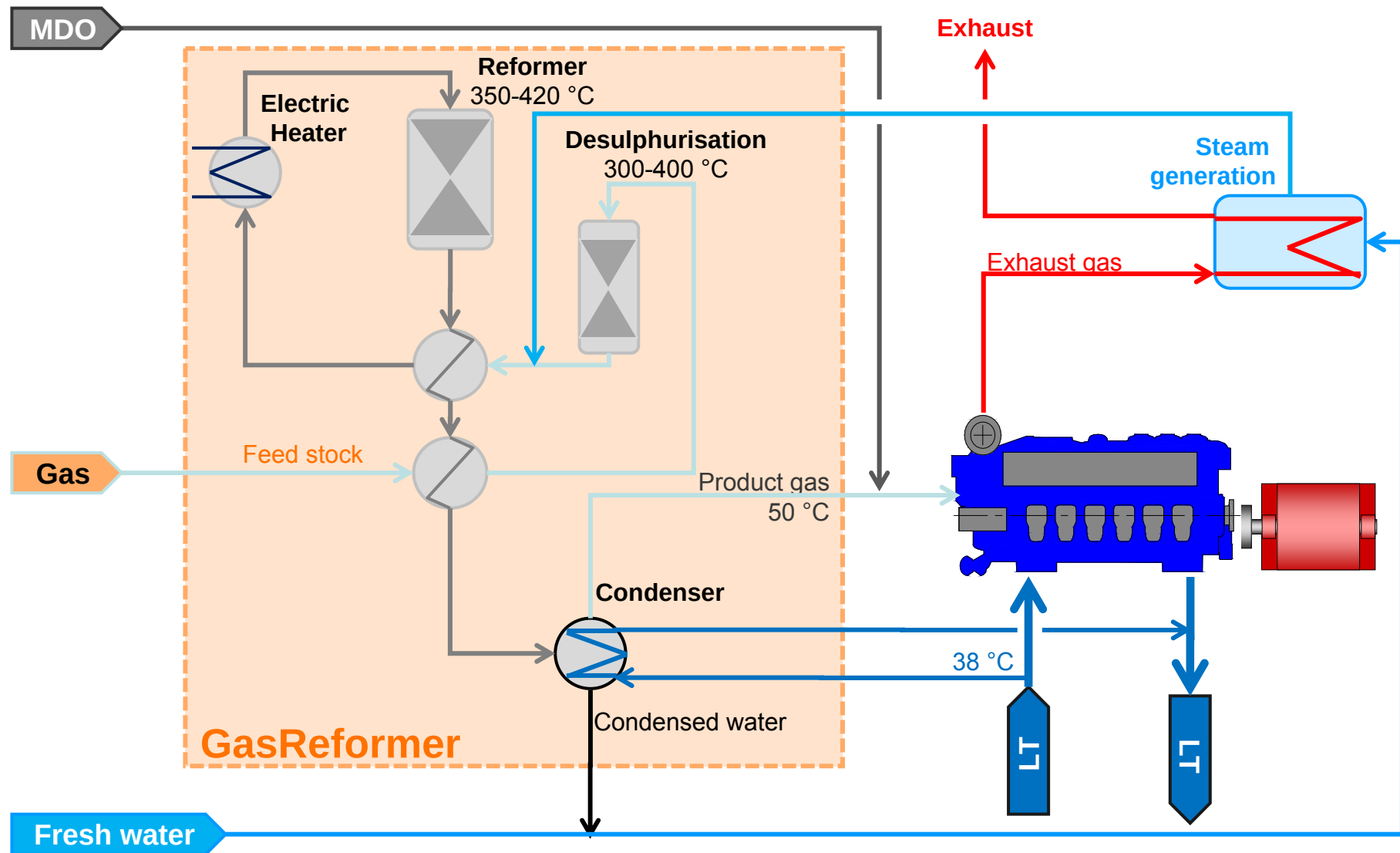
### Reforming parameters

|                 |                     |
|-----------------|---------------------|
| Temperature     | 383 °C              |
| Pressure        | 7,6 barg            |
| Oxygen / carbon | 1,00 mol / 1,00 mol |

Note:

Pressure of inlet gas is 8 barg

## Process of gas reforming and connection with engine



## Main components of gas reformer AES 3000

| TAG                                 | Equipment                               | Specification                            |
|-------------------------------------|-----------------------------------------|------------------------------------------|
| <b>Heat Exchangers and Heaters</b>  |                                         |                                          |
| E100                                | Plate Heat Exchanger                    |                                          |
| E200                                | Plate Heat Exchanger                    |                                          |
| E300                                | Plate Heat Exchanger                    | Location different in GasReformer unit   |
| E400                                | Plate Heat Exchanger (Condenser)        |                                          |
| <b>Reactors, Vessels, and Drums</b> |                                         |                                          |
| R101                                | Desulfurization Vessel 1                |                                          |
| R102                                | Desulfurization Vessel 2                |                                          |
| R103                                | Reformer Vessel                         | Parallel R103a & R103b used for AES 3000 |
| D101                                | Condensate Separator Drum               |                                          |
| D102                                | De-aeration Vessel for Condensate Water |                                          |
| D501                                | Condensate Pot                          | Venting line                             |
| <b>Other Components</b>             |                                         |                                          |
| B101                                | Gas Filter                              |                                          |
| B102                                | Steam Strainer                          |                                          |
| B102                                | Ejector                                 |                                          |
| P401                                | Circulation Blower                      |                                          |

# Heat Exchangers and Heaters

## Heat Exchangers E100 and E200

- The GasReformer has two integrated plate heat exchangers E100 and E200. These heat exchangers are responsible for the heating of the feed streams to desulfurization (250-350 °C) and reformer (350-420 °C) vessels and responsible for cooling of the reformer product gas before entering the Condenser (E400).

## Electrical heater E300

- The electrical heater E300 upstream the reformer vessel is used to control the reformer vessel inlet and outlet temperatures. The set-point temperature for reformer vessel inlet depends on the hydrocarbon composition of the feed gas and thus varies by case. However, the minimum temperature of reformer feed is always 350 °C.

## Heat Exchanger E400 (Condenser)

- In the GasReformer unit product gas line, three plate heat exchangers (E200, E100, and E400) operate in cascade. These heat exchangers take care of the required cooling of the product gas from 350-450 °C to final 50 °C. The plate heat exchangers E200 and E100 are integrated into the GasReformer process, whereas the condenser E400 has low temperature cooling water on its cold side. The last heat exchanger (E400) not only cools down the product gas to 50 °C but also condensates all excess steam.

# Reactors, Vessels, and Drums 1

## Condensate Separator Drum D101

- The excess steam is condensed in the plate heat exchanger E400 and separated from the product gas in the separator drum D101 downstream. The condensed water is collected in the separator drum D101 and recycled to the Steam Generation unit (external). By recycling, the need for fresh water is minimized.
- The product gas is now considered as dry product, still containing 1-2 mol-% of moisture. This cooling and the separation of condensates are sufficient to prevent condensate formation in the Gas Valve Unit (GVU) downstream.

## De-aeration Vessel for Condensate Water D102

- The condensate line from the Separator drum (D101) is equipped with a de-aeration vessel D102 to ensure that gaseous compounds do not end up in the recycled water and the Steam Generation unit that are located outside Hazardous area Zone 2. In case, gaseous compounds are detected in the de-aeration vessel D102, these indicate of process operation failure or leakage and an emergency shut-down (ESD) is automatically activated.

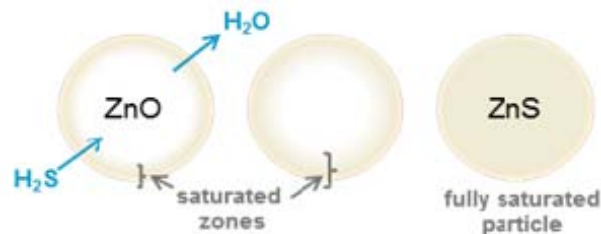
## Condensate Pot D501 in Venting Line

- The gas venting line is equipped with a condensate pot D501 to remove condensate and prevent possible obstructions for normal operation of venting line. The venting line is open to atmosphere and is only used during equipment shut-down or failure mode, such as pressure safety valve opening. Condensate can accumulate in the piping during periods of inactivity.

# Reactors, Vessels, and Drums 2

## Desulfurization Vessels R101 and R102

- In the desulfurization vessels R101 and R102 all hydrogen sulfide (H<sub>2</sub>S) from the feed gas is removed by absorbing H<sub>2</sub>S into a ZnO absorbent:  
$$\text{ZnO} + \text{H}_2\text{S} = \text{ZnS} + \text{H}_2\text{O} \quad (200\text{--}400\text{ }^\circ\text{C})$$
- The desulfurization is designed to be a 2-bed system, where the first vessel R101 consist the main sulfur removal bed. The second vessel R102 acts as guard. The ZnO absorbent bed is saturated in gradients, as presented in Figure below, and the sulfur level measured from first desulfurization vessel R101 outlet indicates of absorbent replacement needs.
- Gradual saturation of ZnO particles



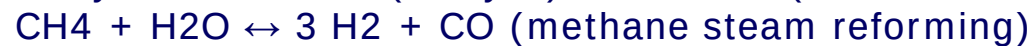
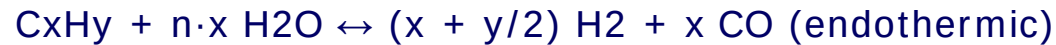
- NOTE: Only in case, where 0 ppmv H<sub>2</sub>S is in feed, one desulfurization vessel is sufficient.
- NOTE: The desulfurization reaction is an equilibrium reaction, and therefore, the feed to desulfurization should not contain moisture (H<sub>2</sub>O < 2 mol-%) that inhibits the H<sub>2</sub>S absorption. The feed gas should not contain more than 2 mol-% of H<sub>2</sub>, if CO<sub>2</sub> is present, as these react together forming H<sub>2</sub>O + CO at this low temperature.
- NOTE: GasReformer unit provides sampling points after both the desulfurization vessels, but analysis tools are not included.

# Reactors, Vessels, and Drums 3

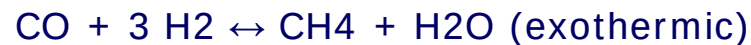
## Reformer Vessel R103

- The reformer vessel R103 is the main module of the GasReformer unit. In the reformer the methane number (MN) of the feed gas is improved by converting all higher hydrocarbons to methane, hydrogen, and carbon dioxide. The main reactions in the reformer are:

### Steam reforming



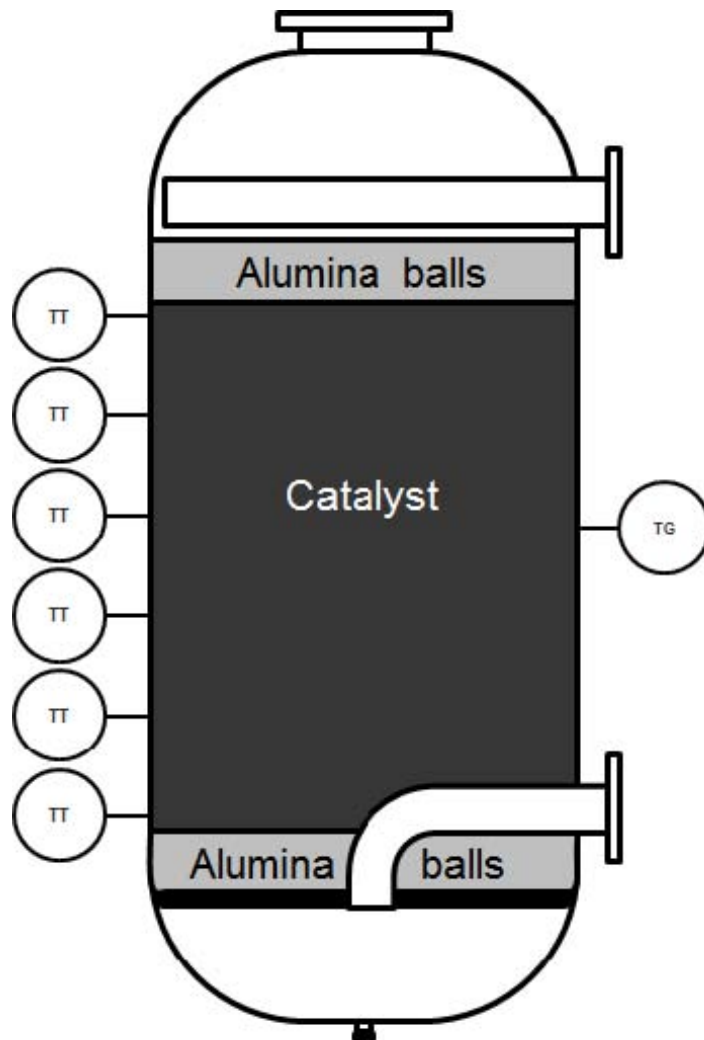
### Methanation



### Water-gas shift



## Cross section of the reformer vessel



These reactions take place on an active nickel catalyst, which is packed as a bed between layers of alumina balls. The upper alumina layer ( $\varnothing 1''$ ) is to avoid turbulence in the top part and to help the distribution of the feed gas. The lower alumina layer ( $\varnothing \frac{1}{2}''$ ) is to hold back dust and particles from the catalyst bed. The cross section of the reformer vessel is shown in Figure on the left.

# Other Components

## Ejector B102

- To maintain high activity of the reformer catalyst the reformer vessel R103 feed should contain 1-2 mol-% hydrogen (H<sub>2</sub>). The small amount of hydrogen in feed keeps the nickel (Ni) sites in reduced, active state.
- As pure hydrogen supply is usually not available, 10-30 % of the hot product gas from reformer vessel outlet is recycled by the ejector B102 to the reformer vessel feed.

## Gas Filter B101 and Steam Strainer B201

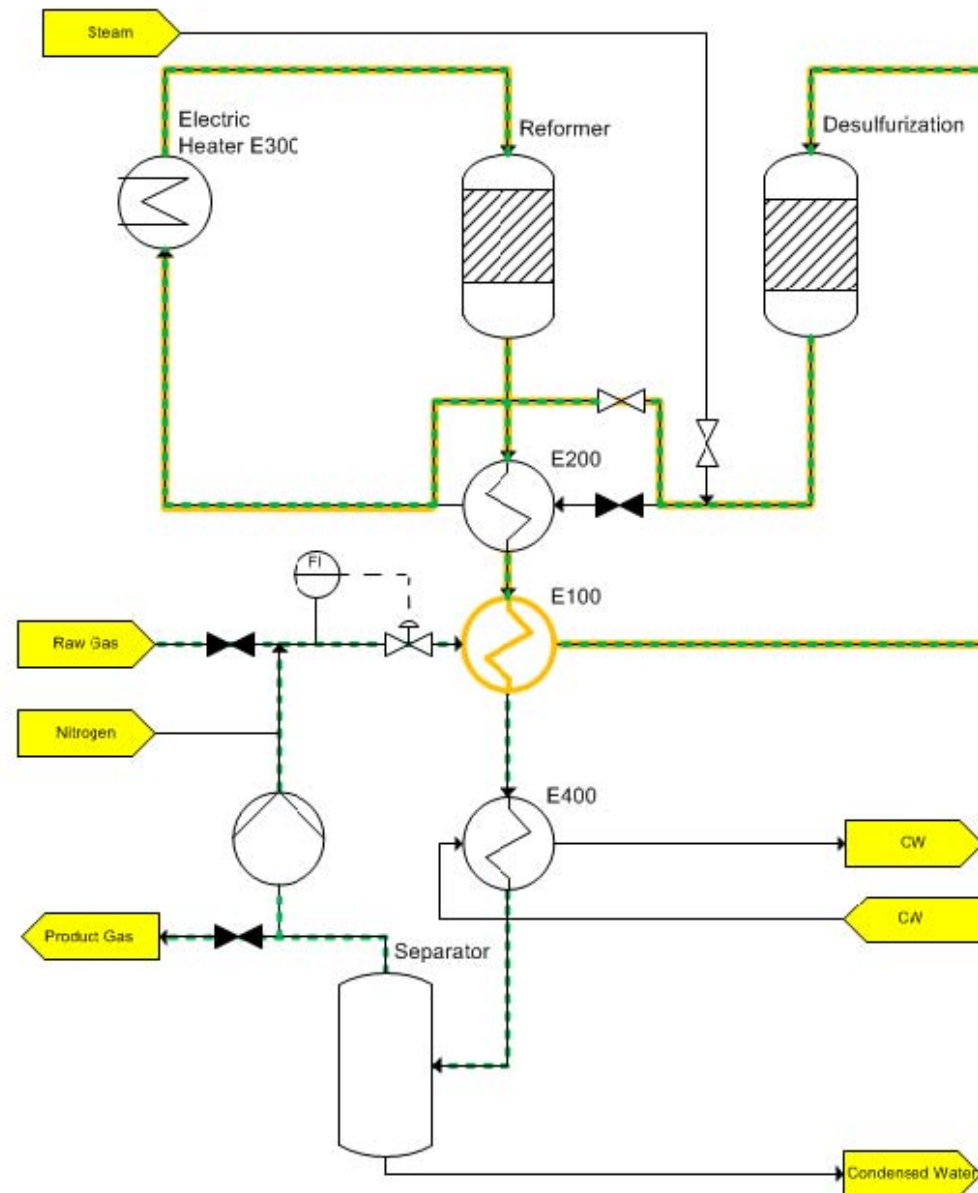
- The gas filter B101 and steam strainer B201 in inlet lines take care that no impurities such as solid matter will enter the process.

## Circulation Blower P401

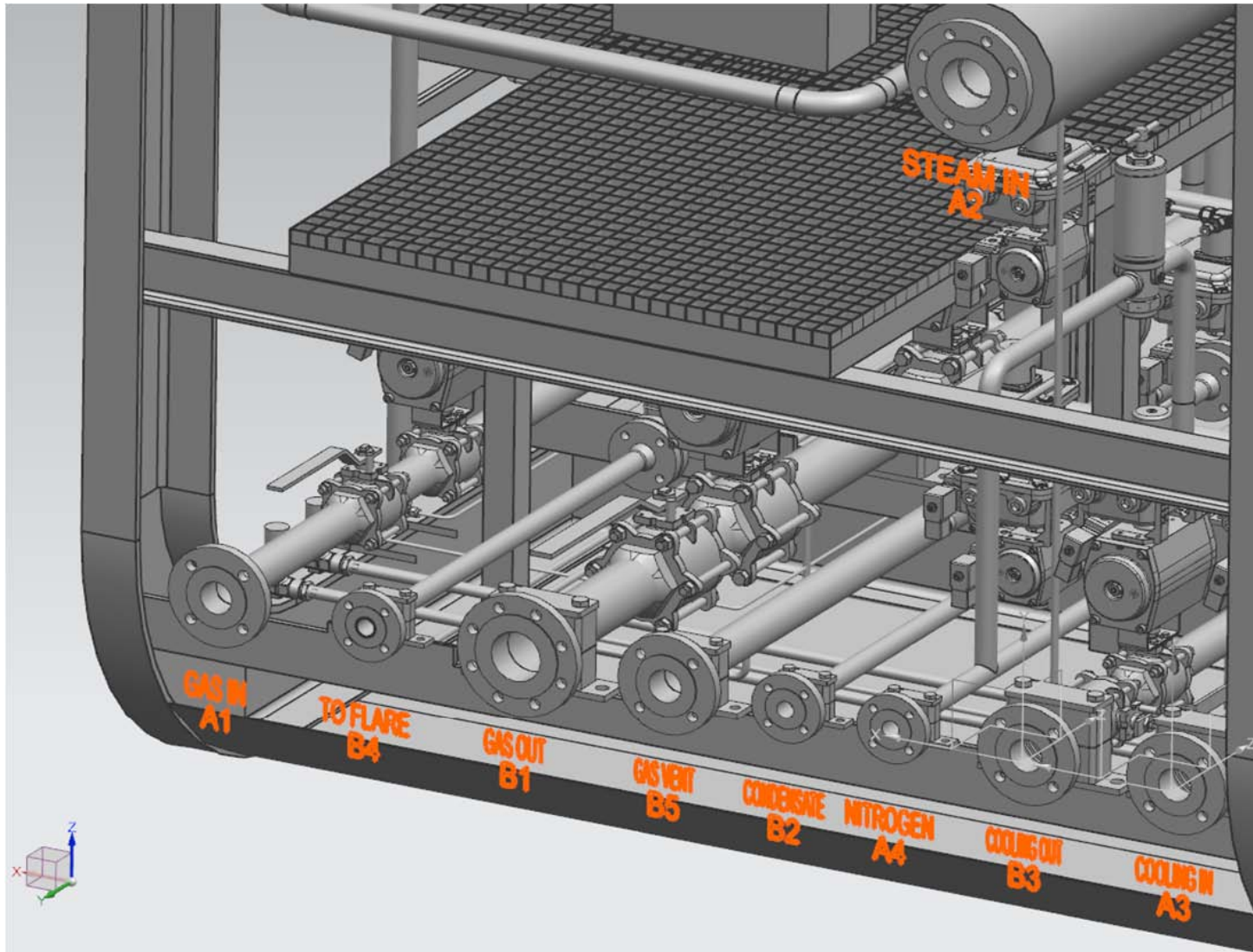
- The blower P401 is used for circulating low pressure nitrogen in the process system. Nitrogen circulation is needed only for heat-up and cooling of the GasReformer unit.

## Use of nitrogen

Nitrogen is consumed in start-up and shut-down of the GasReformer process, as well as in emergency and flushing situations. In the start-up sequence, nitrogen is circulated through the system, and an electrical heater is heating the media, see Figure on the right. Before normal operation nitrogen is vented out of the system and the unit is pressurized with feed gas.

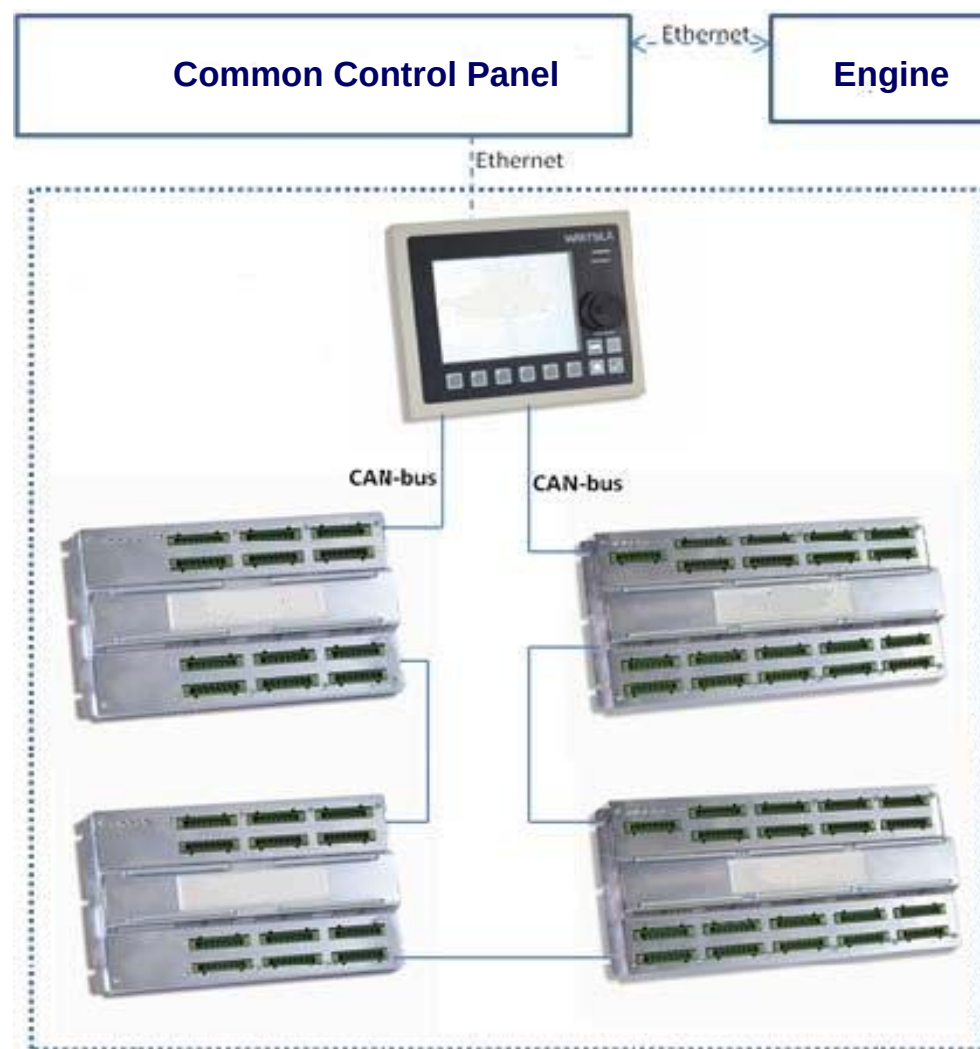


## Interface of process

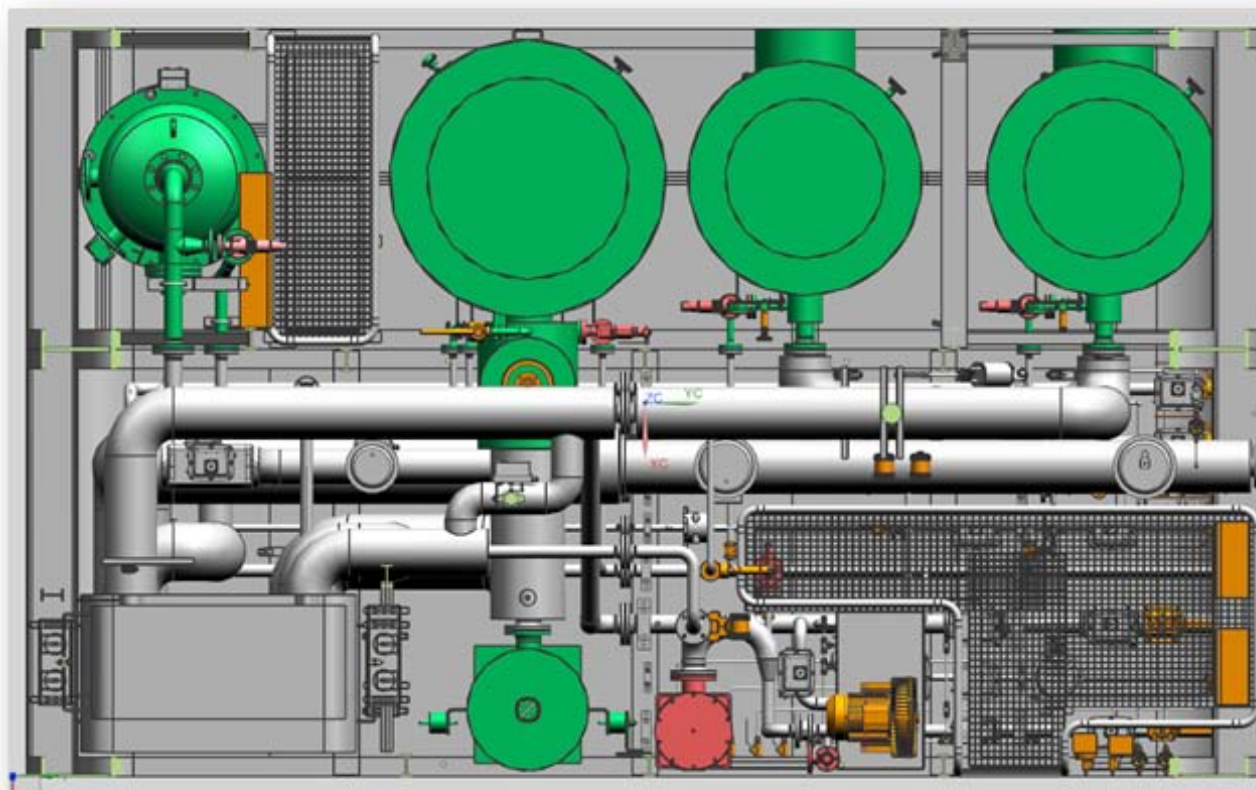


## Automatic control system

**Gas reformer AES3000  
Main Control Panel**



# Maintenance Plan



- Regular maintenance (lifetime related, inspection)
- Access required (manual valves, check etc.)
- Inspection

## Period of operation / service costs

| Service intervals                                                         |                        |
|---------------------------------------------------------------------------|------------------------|
| Number of converting column R103 (or R103a+b)                             | 1 (2)                  |
| Estimated period of catalyst deterioration                                | 3-5 years              |
| column volume                                                             | 6 (4,3) m <sup>3</sup> |
| filled column weight                                                      | 7900 (4300) kg         |
| catalyst weight                                                           | 3900 (1930) kg         |
| Number of Desulfurization Vessels R101+R102                               | 2                      |
| Estimated period of catalyst deterioration (10 ppm H <sub>2</sub> S) R101 | 1 year                 |
| Estimated period of catalyst preventive column R102                       | 3 years                |
| Colimn volume                                                             | 1.6 m <sup>3</sup>     |
| filled column weight                                                      | 2800 kg                |
| catalyst weight                                                           | 1540 kg                |
| Filtrs lifetime                                                           | ~ 1 year               |

**Service costs for 15 years of gas reformer AES 3000 operation are about 120 thousand Euro per year (0.75 Euro / MW) and include the replacement of catalysts, components and working hours.**

# Period of operation / consumable materials

| Consumable materials           |            |
|--------------------------------|------------|
| Water / Steam                  |            |
| Steam                          | 3989 kg/h  |
| Steam boiler condensed water   | 3012 kg/h  |
| Steam boiler consumed water    | 977 kg/h   |
| Cooling water                  |            |
| Inlet temperature              | 38°C       |
| Outlet temperature             | 98°C       |
| Steam                          | 46569 kg/h |
| Enthalpy                       | 2908 kW    |
| Nitrogen                       |            |
| Pressure                       | 9 barg     |
| Purity                         | 99,9%      |
| Consumption at start           | 6 nm3      |
| Consumption at cleaning / stop | 27 nm3     |
| Own consumption of electricity |            |
| Operation at par               | 111 kW     |
| Start                          | 50 kW      |
| Stop                           | 20 kW      |

## Contact

### *AESI AG*

Austro Energy Systems  
International  
Produktions & Beteiligungs AG

**Bräunerstraße 10, Top 5  
1010 Vienna  
Austria**

**Contacts:**

**+43 1 535 17 70**

**Fax:**

**+43 1 535 17 70 32**

**E-mail:**

**office@aes-int.com, sec@aes-int.com**

**Web site: [www.aes-int.com](http://www.aes-int.com)**

